

CanSat Project Presentation

Innovating in miniature satellite technology for exploration and education.



What is CanSat?

A CanSat is more than just a model; it's a fully functional satellite simulation, meticulously engineered to fit within the confines of a standard soda can.

Miniature Satellite

A CanSat integrates essential satellite subsystems like power, sensors, and communication into a compact, soda-can-sized package.

Integrated Payload

It houses a scientific payload designed to perform specific experiments during its descent.

Descent & Landing

Once launched, it deploys a parachute to control its fall, collecting data before a safe landing.

Uses and Mission of CanSat

CanSats serve as versatile platforms for both scientific inquiry and educational development.

Atmospheric Data Collection

Gathers crucial atmospheric data, including temperature, pressure, altitude, and GPS location.



Secondary Experiments

Capable of conducting additional experiments like soil sampling or chemical analysis.

Technology Development

Aids in the rapid prototyping and testing of miniature satellite technologies.



Educational Outreach

An invaluable tool for STEM education, inspiring future engineers and scientists.

How CanSat Works

Our CanSat follows a precise operational sequence from launch to data transmission.

1

Launch

Deployed from a drone or small rocket, reaching altitudes of approximately 20 meters.

2

Parachute Deployment

A parachute automatically deploys, ensuring a controlled and safe descent.

3

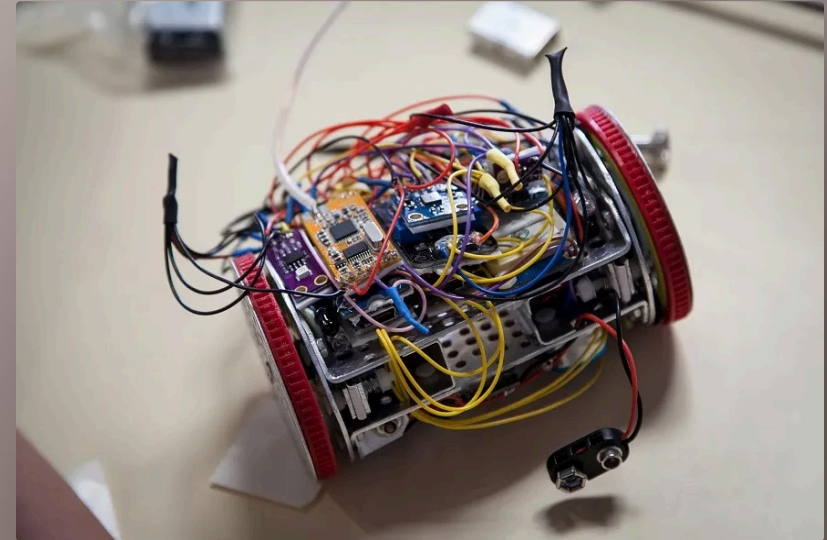
Data Transmission

Onboard sensors continuously collect and transmit real-time data to our ground station.

Key Components of Our CanSat

Our CanSat is engineered with precision components to ensure robust performance and reliable data collection.

- **Advanced Sensors:** Integrated GPS for location tracking, barometers for pressure and altitude, an accelerometer and magnetometer for orientation, and highly accurate temperature sensors.
- **Central Control System:** A powerful microcontroller efficiently manages all data acquisition, processing, and transmission.
- **Efficient Power System:** A lightweight yet high-capacity LiPo battery provides stable power to all electronic components throughout the mission.



Scientific Payload and Experiments

Our CanSat's primary mission is focused on precise environmental data acquisition and real-time visualization.

Primary Mission

The CanSat's core task is to measure and transmit air temperature and pressure data every second during its descent. This continuous stream provides a detailed atmospheric profile.

Data Visualization

Collected data is instantly relayed to our ground station and visualized live through a custom GUI, featuring interactive graphs and maps for immediate analysis.



Mechanical and Electrical Design Highlights

Our project emphasizes precision engineering in both its mechanical and electrical design, ensuring optimal performance.

1 Compact PCB Design

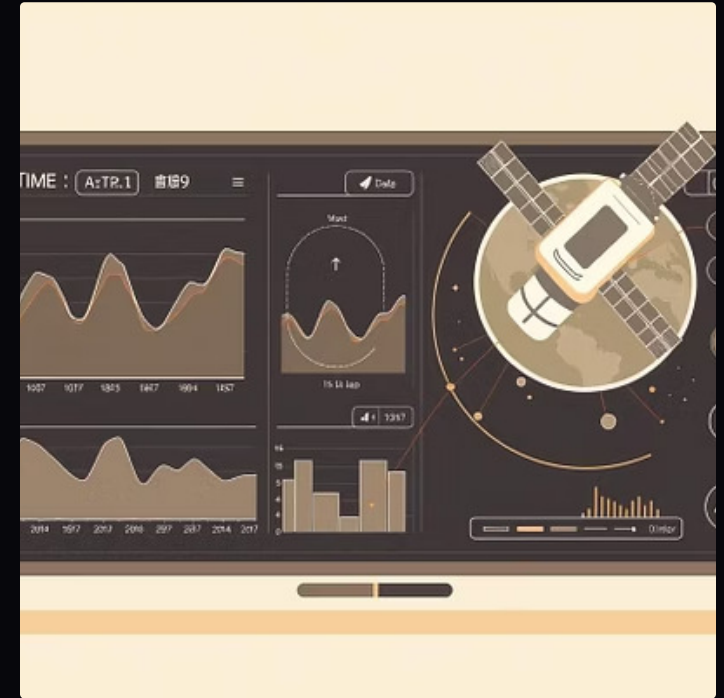
We meticulously designed and hand-assembled a compact Printed Circuit Board, integrating all sensors and components onto a single, efficient platform.

2 Custom 3D Printed Parts

Utilizing advanced 3D printing, we created custom parts for secure housing and optimal aerodynamic stability, enhancing flight performance.

Our CanSat in Action

Experience the visual journey of our CanSat project through a collection of our project videos and images, showcasing its development and missions.



Meet Our Team

The dedicated individuals who brought the CanSat project to life.

Leader

Chakrapani

Senior Guide

Vedansh

Team Members

- Bhav Makija
- Ananya Sinha
- Manish
- Mohit
- Pavan Kalyan
- Sathvik